

Name: Solutions (please print)

Signature: _____

ECE 2202 – Quiz 4

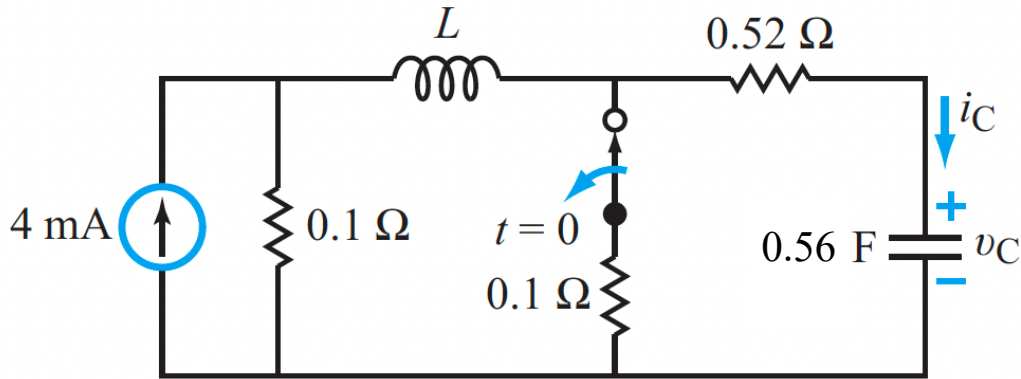
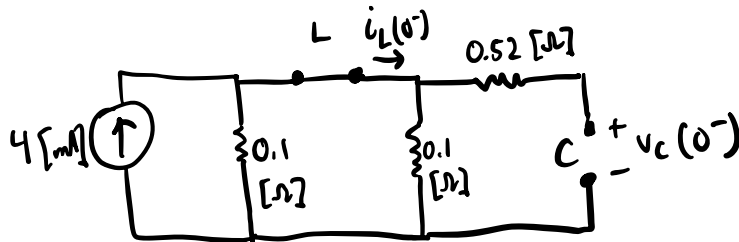
October 23, 2025

1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.
2. Show all units in solutions and figures. Units in the quiz will be included between square brackets.
3. You will have 40 minutes to work on the quiz.
4. Write your ID number on every page that you show your work.

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Problem 1 (10 pts)

Choose the value of the inductor in the circuit below so that v_C exhibits a critically damped response and determine $v_C(t)$ for $t \geq 0$.

 $t = 0^-$ 

CDR gives $i_L(0^-) = 2 \text{ [mA]}$
 $v_C(0^-) = 2 \text{ [mA]} \cdot 0.1 \text{ [Ω]} = 0.2 \text{ [mV]}$

 $t = 0$

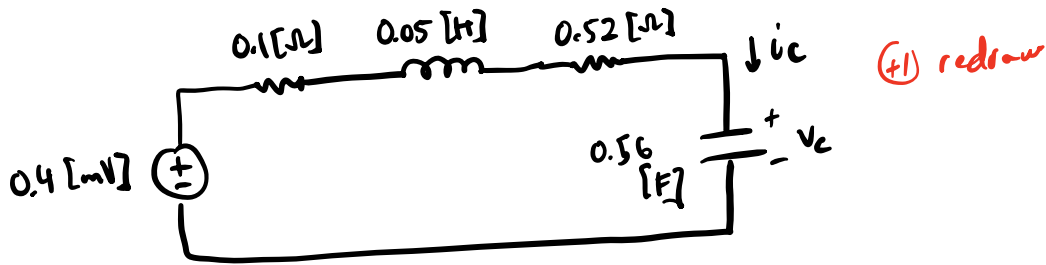
$$i_C(0) = i_L(0) = i_L(0^-) = 2 \text{ [mA]}$$

$$v_C(0) = v_C(0^-) = 0.2 \text{ [mV]}$$

$$v_C'(0) = \frac{i_C(0)}{C} = \frac{2 \text{ [mA]}}{0.56 \text{ [F]}} = 3.6 \text{ [mV]}$$

(+1) correct values
initial
conditions

* a source transformation will lead to a series RLC configuration
 (+1) next page

$t > 0$ 

damping parameters: (+1) correct values

$$\alpha = \frac{R}{2L} = \frac{0.52 + 0.1 \text{ [}\Omega\text{]}}{2(0.05 \text{ [H]})} = 6.2 \text{ [s}^{-1}\text{]}$$

 $\alpha > \omega_0 \Rightarrow \text{overdamped}$

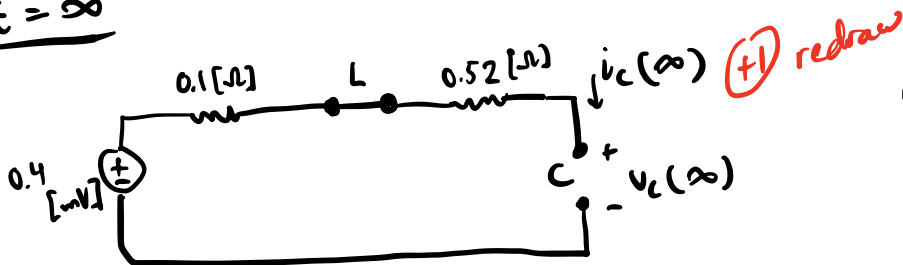
$$\omega_0 = \frac{1}{\sqrt{LC}} = 6 \text{ [rad/s]}$$

general solution:

$$v_c(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_c(\infty)$$

go to table 6-2 for roots s_1 + s_2 :

$$s_1 = -4.64, s_2 = -7.76 \quad (+1) \text{ correct values roots}$$

 $t = \infty$ 

$$v_c(\infty) = 0.4 \text{ [mV]}$$

(+1) correct steady state value

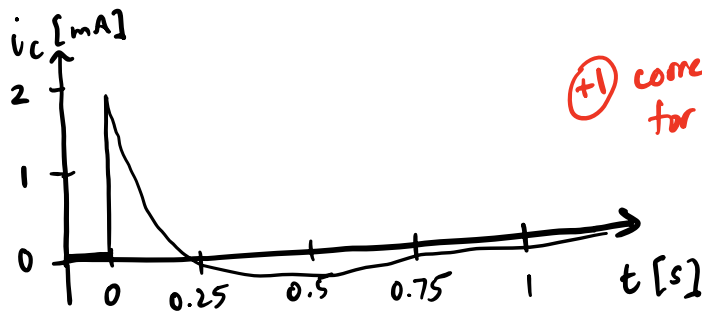
go to table 6-2 for A_1 + A_2 (we found $v_c'(0)$ above):

$$A_1 = 0.66 \text{ [mV]}, A_2 = -0.86 \text{ [mV]}$$

$$v_c(t) = [0.4 + 0.66 e^{-4.64t} - 0.86 e^{-7.76t}] \text{ [mV]}$$

(+2) final equation $v_c(t)$ + $i_c(t)$

$$i_c(t) = C v_c'(t) = [-1.7 e^{-4.64t} + 3.7 e^{-7.76t}] \text{ [mA]} \text{ for } t \geq 0$$

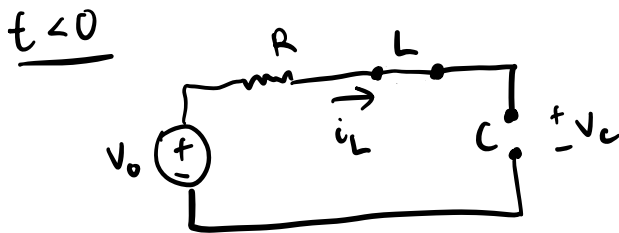
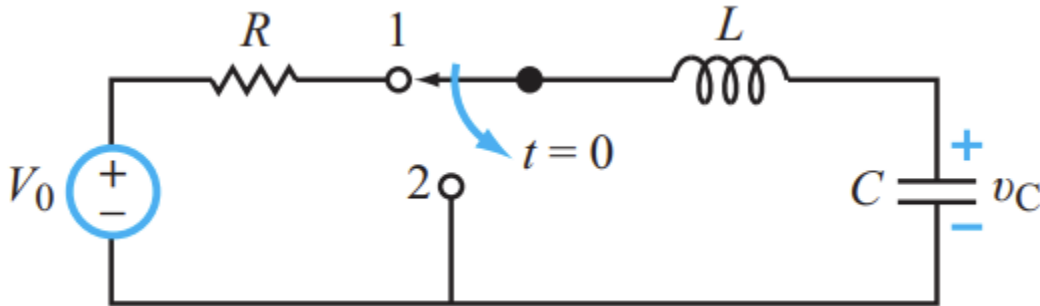
Plot $i_c(t)$:

(+1) correct shape for plot (no values need to be marked)

Problem 2 (10 pts)

$$V_0 = 5 \text{ [V]}, R = 1 \text{ [k}\Omega\text{]}, L = 3.2 \text{ [H]}, C = 0.2 \text{ [\mu F]}$$

In the circuit below, the switch is moved from position 1 to position 2 at $t = 0$.
Provide an expression for $v_C(t)$ for $t \geq 0$.

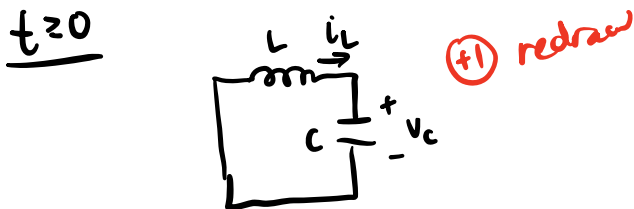


initial conditions:

$$i_L(0) = 0 \text{ [A]}$$

$$v_C(0) = 5 \text{ [V]}$$

(+1) correct values
initial conditions



(+1) redraw

series LC with no R or source $\Rightarrow$ damping parameters

$$\alpha = \frac{R}{2L} = 0$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 1250 \text{ [rad/s]}$$

$$\alpha < \omega_0 \Rightarrow \text{underdamped}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 1250 \text{ [rad/s]}$$

general solution for underdamped:

$$v_C(t) = e^{-\alpha t} [D_1 \cos \omega_d t + D_2 \sin \omega_d t]$$

next page

$$v_c(t) = D_1 \cos 1250t + D_2 \sin 1250t$$

go to table 6-2 for D_1 : (+1) correct value D_1

$$D_1 = 5 \text{ [V]} = v_c(0)$$

for D_2 : (+2) correct value D_2

$$\frac{dv_c(t)}{dt} = -\omega_d D_1 \sin \omega_d t + \omega_d D_2 \cos \omega_d t$$

$$i_L(t) = C \cdot \frac{dv_c(t)}{dt}$$

$$i_L(0) = C \cdot \frac{dv_c(0)}{dt} = 0.2 \text{ [uF]} \cdot (-1250 \cdot D_1 \cdot 0 - 1250 \cdot D_2 \cdot 1)$$

$$i_L(0) = 0$$

$$\rightarrow 0.2 \text{ [uF]} \cdot (-1250 \cdot D_1 \cdot 0 - 1250 \cdot D_2 \cdot 1) = 0$$

$$D_2 = 0$$

$$v_c(t) = 5 \cos 1250t \quad \text{for } t \geq 0$$

(+2) correct equation $v_c(t)$

$$i = \frac{dq}{dt} \quad v = \frac{dw}{dq} \quad p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = iv$$

$$i = C \frac{dv}{dt} \quad v = L \frac{di}{dt}$$

$$v = iR$$

$$R_1 || R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$p = Cv \frac{dv}{dt} \quad p = Li \frac{di}{dt}$$

$$p = iv = i^2 R = \frac{v^2}{R}$$

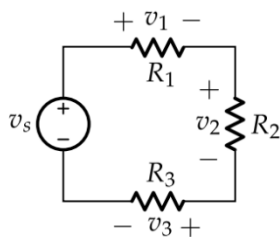
$$w = \frac{1}{2} Li^2 \quad w = \frac{1}{2} Cv^2$$

$$R = \frac{v}{i}$$

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)] e^{-(t-T_0)/\tau}$$

$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)] e^{-(t-T_0)/\tau}$$

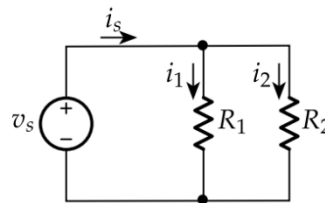
(Switch action at $t = T_0$)



$$v_1 = \frac{R_1}{R_1 + R_2 + R_3} v_s$$

$$v_2 = \frac{R_2}{R_1 + R_2 + R_3} v_s$$

$$v_3 = \frac{R_3}{R_1 + R_2 + R_3} v_s$$



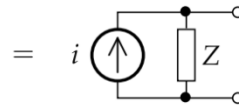
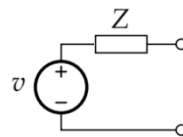
$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$

$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

$$v_B = \underbrace{V_B}_{\text{DC}} + \underbrace{v_b}_{\text{AC}}$$

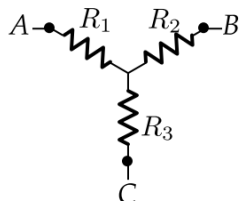
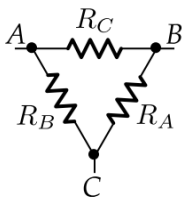
$\bar{V}_b \rightarrow$ Phasor notation

$$\text{CMRR} = 20 \log_{10} \frac{|A_d|}{|A_{cm}|}$$



$$v = i \cdot Z$$

$$i = \frac{v}{Z}$$



$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$

Table 6-2: General solution for second-order circuits for $t \geq 0$.

$x(t)$ = unknown variable (voltage or current)	
Differential equation:	$x'' + ax' + bx = c$
Initial conditions:	$x(0)$ and $x'(0)$
Final condition:	$x(\infty) = \frac{c}{b}$
$\alpha = \frac{a}{2}$	$\omega_0 = \sqrt{b}$
Overdamped Response $\alpha > \omega_0$	
$x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)] u(t)$	
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$
$A_1 = \frac{x'(0) - s_2[x(0) - x(\infty)]}{s_1 - s_2}$	$A_2 = -\left[\frac{x'(0) - s_1[x(0) - x(\infty)]}{s_1 - s_2} \right]$
Critically Damped $\alpha = \omega_0$	
$x(t) = [(B_1 + B_2 t)e^{-\alpha t} + x(\infty)] u(t)$	
$B_1 = x(0) - x(\infty)$	$B_2 = x'(0) + \alpha[x(0) - x(\infty)]$
Underdamped $\alpha < \omega_0$	
$x(t) = [(D_1 \cos \omega_d t + D_2 \sin \omega_d t)e^{-\alpha t} + x(\infty)] u(t)$	
$D_1 = x(0) - x(\infty)$	$D_2 = \frac{x'(0) + \alpha[x(0) - x(\infty)]}{\omega_d}$
$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$	
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \frac{1}{\sqrt{LC}}$
	$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$$\vec{S} = \overrightarrow{V_{rms}} \overleftarrow{I_{rms}}^* = |\overrightarrow{I_{rms}}|^2 \vec{Z} = P + jQ$$

$$P = \frac{V_m I_m}{2} \cos \theta$$

$$Q = \frac{V_m I_m}{2} \sin \theta$$

$$pf = \cos \theta$$

$$rf = \sin \theta$$